

How did COVID-19 Affect Playlist Followers on Spotify? *

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Abstract

The COVID-19 pandemic disrupted traditional music consumption and accelerated the shift toward streaming platforms. While global subscriptions grew, less is known about how engagement changed within platforms. This study examines weekly changes in Spotify playlist followers from October 2019 to October 2020 to assess shifts in user behavior and stakeholder visibility. We estimate a unit-fixed effects model using three measures of pandemic impact: the official declaration date, a government stringency index, and residential mobility data. We find no significant change in average follower growth after the pandemic declaration, but growth increased when users spent more time at home. Playlists with moderate or small existing followings experienced stronger growth than top-tier (i.e., most popular) playlists, suggesting engagement expanded across a wider range of content. Spotify-curated playlists were more resilient and productive than those curated by major labels or independent entities, even after accounting for platform-driven promotion. While playlists with a higher share of major label content saw some growth, track-level popularity had a stronger effect. These findings highlight how behavioral shifts affected existing music content consumption patterns and stakeholders' visibility within the streaming platforms.

Keywords: Playlist Followers; Lockdown Restrictions; Music Streaming; Pandemic; Playlist Featuring

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1 Introduction

After the outbreak of the COVID-19 pandemic, the music industry, along with many creative industries suffered from government interventions and restrictions on live performance and in-person gatherings (WIPO, 2022). These restrictions, along with reduced commuting and social activities, contributed to a decline in music consumption and streaming activity, particularly during the first months following the outbreak in early 2020 (Denk et al., 2022; Sim et al., 2022). As a result, revenues from both live concerts and physical and digital record sales dropped significantly, at least 18% in the US and 5% globally (RIAA, 2021; IFPI, 2021). In contrast, music streaming platforms reported growth in both user numbers and subscription revenues during and after the pandemic at least 13% in the US and 18% globally (RIAA, 2021; IFPI, 2021). This contradiction suggests that the entry of new users might affect how existing music content was consumed and how stakeholders within the platform gained visibility and influence, especially from the perspective of audience behavior and engagement.

As the pandemic pushed audiences away from traditional formats more toward digital platforms, streaming services became the primary channel for music discovery and consumption. These platforms generally provide users with on-demand access to extensive music catalogs and support user engagement through features like algorithmic recommendations and curated playlists. These curated playlists are important in shaping how listeners discover and engage with music, as they introduce users to a wider variety of tracks and facilitate music exploration (Datta et al., 2018). Moreover, their follower counts are highly correlated with actual active listeners (Datta & Pachali, 2024). Since playlists are both discovery tools and exposure channels, changes in their follower numbers offer insight into how users’ behavior evolved during the pandemic and how that behavior may have affected the visibility and positioning of different stakeholders within the platform.

The music streaming ecosystem is composed of stakeholders with vastly different levels of influence and resources, from global platforms and major record labels to independent curators and artists. Given their varying resources and market power, stakeholders’ abilities to counteract the adverse effects of the pandemic or capitalize on new opportunities likely varied. For instance, playlists curated by streaming platforms might have been able to leverage algorithmic promotion and user data to retain followers, while major label-backed playlists could have gained traction as traditional promotion channels became more limited. Independent curators, however, may have faced greater challenges in maintaining visibility and engagement. Building on this, our research investigates the changes in playlist followers, an observable proxy for sustained user engagement on the platform, across different curators before and after the pandemic declaration. We are particularly interested in understanding whether certain types of playlists were more resilient than others in maintaining or even gaining followers during the pandemic. Thus, we examine how playlist follower changes varied not only across curators but also across different levels of playlist popularity and content composition, as these characteristics are likely to moderate the

extent to which playlists retained or gained followers during the pandemic. Accordingly, we pose the following empirical research questions:

RQ 1: How did the number of followers for playlists change during the pandemic?

RQ 2: How did follower dynamics differ across playlists with varying levels of popularity during the pandemic?

RQ 3: How did playlists curated by music streaming platforms, major labels, and independent curators differ in their resilience to changes in follower numbers during the pandemic?

RQ 4: Did playlists featuring a higher proportion of tracks from major labels experience greater follower growth during the pandemic?

These research questions aim to uncover shifts in user engagement patterns via playlists following and competitive positioning within the music streaming platform during the pandemic. By examining overall changes in playlist followers (RQ1), we assess how the pandemic influenced listener engagement across playlist types, providing insight into platform performance and potential shifts in exposure through playlists. Investigating the impact on playlists with different levels of popularity¹ (RQ2) helps determine whether the pandemic intensified or narrowed disparities between highly followed and niche playlists. Examining variation in resilience across curators (RQ3), who differ in their resources and promotional capacity, sheds light on how stakeholders managed follower retention and growth during the pandemic. Finally, evaluating whether playlists with a higher share of major label content gained more followers (RQ4) reveals how established industry players may have sustained or strengthened their influence over streaming behavior. Together, these questions provide insights into how the pandemic may have influenced the relative visibility and engagement outcomes for different stakeholders, with potential implications for platform dynamics and industry structure.

To answer our research questions, we estimate weekly changes in Spotify playlist followers before and after the COVID-19 pandemic declaration, focusing on shifts in overall follower counts across playlists with different popularity levels, curated by different curators, and having varying proportions of major label content. We utilize playlist information from Spotify covering playlists from October 2019 to October 2020, sourced from *Chartmetric.com* for playlist-level follower data and *Everynoise.com* for playlist feature information. The latter feature data captures instances when playlists are promoted on the Spotify user interface (i.e. Browse or Search Page). Controlling for this promotional information enables us to distinguish organic follower growth from gains driven by platform-driven exposure. We employ a unit fixed effects model to control for the time-invariant characteristics of playlists and account for pandemic-related variation (Papies et al., 2022; Papies & van Heerde, 2017) using three complementary measures: a binary indicator for the official pandemic declaration date, a country-weighted government stringency index, and a

¹In the music industry literature, highly popular playlists are often linked to the broader superstar phenomenon (Rosen 1981; Elberse 2008). We use the term top-tier playlists to remain consistent with our empirical categorization while drawing conceptually on this literature in Section 2.2.

country-weighted residential mobility index. The first measure captures the average shift in playlist engagement following the initial pandemic declaration, while the other two reflect the evolving impact of government restrictions and public mobility patterns. Together, these indicators help us assess how varying lockdown measures and behavioral shifts influenced user engagement across different types of playlists and stakeholders.

Our findings suggest structural shifts in playlist follower engagement during the COVID-19 pandemic, offering empirical insights into how user behavior and stakeholder visibility evolved. Although we found no significant change in the average rate of follower growth after the pandemic declaration (RQ1), follower growth responded positively to increased time spent at home and stricter government restrictions, implying that playlist engagement was more sensitive to behavioral shifts than to the pandemic event itself. We found that follower changes varied substantially across popularity levels (RQ2). Playlists from upper mid-range in the top 5% and lower gained relatively stronger growth, while “superstar” playlists in the top 0.1%, which account for over 50% of all followers, did not experience additional gains. This indicates that follower growth during the pandemic was less concentrated at the top, leading to a more even distribution of engagement across playlists. We also observe significant differential effects across curator type (RQ3). Spotify-curated playlists were evidently more resilient to pandemic-related shifts than those curated by other curators. This advantage held even after accounting for platform-featured promotion, pointing to Spotify’s strong position as both platform and curator. Finally, while playlists with a higher share of major label content did show some positive association with follower growth, the effects were relatively small and ultimately overshadowed by track-level popularity (RQ4). Once we accounted for how frequently tracks appeared across playlists, the role of label affiliation diminished. This suggests that users engaged more with recognizable tracks than with brand-based cues.

The rest of the paper is organized as follows. Section 2 reviews prior research on music streaming platforms, playlist influences, and the impact of the COVID-19 pandemic. This section highlights research constructs and gaps that motivate our analysis. Section 3 introduces the empirical context of this study, describes the dataset, and details the variable operationalization used to answer our research questions. Section 4 details our identification strategy and presents estimation results from a unit fixed effects model, focusing on differences in playlist follower changes across curator types, popularity tiers, and content composition. Section 5 synthesizes the main findings, draws out implications for platforms and curators, and discusses limitations and directions for future research.

2 Background and Conceptual Framework

Previous literature in marketing has long examined how crises affect consumer behavior and firm outcomes. Studies on product-harm crises emphasize firm- or product-specific faults that trig-

ger blame attribution and recovery strategies (Cleeren et al. 2017; Khamitov et al. 2020). The COVID-19 pandemic, however, differs fundamentally from these contexts: it was a systemic, exogenous disruption without a culpable firm, simultaneously affecting entire industries. Recent articles highlight how pandemics reshape marketing more broadly, often in retailing and marketing strategies implementation. For example, Das et al. (2021) discusses how COVID-19 affected the marketing mix and opened new research opportunities related to consumer behavior, branding, and digital platforms. Roggeveen and Sethuraman (2020) highlights pandemic-induced changes in retailing practice, such as the acceleration of online grocery shopping, shifts in consumer preferences, altered demand for luxury versus value-oriented brands, and the redesign of in-store experiences and employee safety measures. Although this literature stream sheds light on pandemic-related changes in retailing strategies and consumer behavior, it does not directly extend to platform-based contexts. Some implications can be drawn. For instance, playlist curation in streaming may be seen as analogous to product strategy in retailing. Yet the nature of consumption differs. While retailing focuses on transactions, streaming engagement depends on ongoing visibility and following, which shape exposure opportunities for artists and curators. As such, existing retail-oriented insights provide only limited guidance for analyzing how stakeholders in music streaming were affected.

This limitation leads to the need for studies that complement marketing and retailing perspectives by focusing on engagement within digital platforms. Prior studies have highlighted the importance of playlists in shaping music consumption and artist visibility. Datta et al. (2018) emphasized how streaming platforms drive variety and demand, while Aguiar & Waldfogel (2021) and Pachali & Datta (2024) demonstrated a strong link between playlist followers, streaming activity, and music discovery. However, most of this work draws on broader platform settings, offering limited empirical insight into how playlist engagement shifts under external shocks, such as the COVID-19 pandemic, which disrupted typical consumption patterns, shifting listening away from traditional formats and toward more reliance on digital platforms. Although COVID-19 has affected music consumption, its influence on playlist followers has received limited attention. Table 1 summarizes relevant studies examining music consumption across formats, with a focus on five aspects related to our research questions: the unit of analysis, dependent variable, consumption formats, stakeholder focus, and whether the study considers pandemic-related effects. These aspects help frame our empirical reasoning and highlight gaps in existing research.

Relevant studies have typically analyzed songs or albums (unit of analysis) using outcomes such as play counts or revenue (dependent variables), within a single consumption format or across multiple formats. In contrast, we focus on playlists and use playlist followers as a measure of user engagement and visibility within streaming platforms. Although some existing work includes curators or labels as stakeholders, few studies differentiate how various curator types, including platforms, major labels, or independent entities, performed within music streaming platforms under pandemic conditions. Similarly, artists and smaller curators remain underexplored, espe-

cially regarding how playlist composition shaped exposure during this period. Finally, while some studies take place during the COVID-19 period, few directly assess how the pandemic influenced follower dynamics, whether through an aggregate shift (RQ1), variation by playlist popularity (RQ2), differential curator resilience (RQ3), or content-level effects (RQ4).

Study	Unit of Analysis	DV (Dependent Variable)	Format	Stakeholder	Pandemic Effect
Papies & Van Heerde (2017)	Recorded Music, Concerts	Sales, Revenue	Physical, Digital	Labels	-
Datta et al. (2018)	Songs	Play counts, Variety	Streaming	-	-
Yeung (2020)	Songs	Streaming counts	Streaming	-	x
Boughanmi & Ansari (2021)	Albums	Success score	Digital, Physical	Labels, Artists	-
Aguiar & Waldfo-gel (2021)	Songs	Streaming counts	Streaming	Curators	-
Sim et al. (2022)	Songs	Streaming counts	Streaming	-	x
Denk et al. (2022)	Music Listeners	Time & money spent on music	Streaming, Physical	-	x
Pachali & Datta (2024)	Playlists	Followers	Streaming	Curators, Labels	-
This Study	Playlists	Followers	Streaming	Labels, Curators, Artists	x

Table 1: Summary of relevant literature on music industry and the effect of COVID-19 pandemic

By focusing on playlist-level changes, accounting for different effects across playlists’ popularity, curator types, and content characteristics, in response to variation in pandemic-specific indicators, this study provides empirical insight into how COVID-19 reshaped platform engagement via playlist consumption and subsequent stakeholder visibility. To guide the analysis, the following four subsections revisit each of the research questions introduced earlier, situating them within the relevant literature and outlining their empirical motivation.

2.1 Overall Changes in Playlist Followers

First, we examine whether playlist engagement, which is captured by follower counts, declined, increased, or remained stable during the pandemic.

Previous research shows that streaming platforms such as Spotify experienced significant shifts in usage during the COVID-19 pandemic, largely driven by changes in users’ daily routines (Sim et al., 2022; Denk et al., 2022). In particular, studies find that audio consumption, especially music streaming, declined in the early phase of the pandemic, likely due to reduced commuting and other mobile activities that typically accompany music listening. These patterns suggest a potential decrease in music engagement across all consumption formats.

Since playlist followers are strongly associated with listener engagement (Aguilar & Waldfogel, 2021), a decline in streaming activity may also indicate lower follower growth, or even a loss of followers, for many playlists. Yet, the number of music streaming subscribers continued to rise globally in 2020 (MIDia Research, 2024), raising the possibility that users remained on platforms despite consuming less content. This disconnect between user base growth and actual usage may stem from behavioral inertia or subscription stickiness (Rietveld & Schilling, 2020), complicating expectations about average engagement metrics, particularly in terms of playlist followers..

Given this contradictory evidence, which is declining usage but rising subscriptions, it remains unclear what happened to playlist followers on average. Users may have consumed less content while retaining their subscriptions, highlighting the need for a longitudinal empirical analysis to uncover actual patterns of engagement during the pandemic. This leads to our first research question: *How did the number of followers for playlists change during the pandemic?*

2.2 Playlist Popularity and the Superstar Effect

We next explore whether the pandemic reshaped the distribution of engagement across playlists with different levels of popularity, from highly followed playlists, which we empirically categorize as “top-tier” but can be conceptually linked to the well-known “superstar” phenomenon in the music industry (Rosen, 1981; Elberse, 2008), to lesser-known long-tail ones.

On streaming platforms, the highly skewed concentration of playlist engagement is evident in playlist follower distributions. A small number of highly popular playlists, which are typically curated by platforms or major labels, attract the bulk of followers, while the majority of playlists make up a long tail of niche, less-followed content (Aguilar et al., 2021; Pachali & Datta, 2024). This reflects the broader “superstar effect,” a phenomenon where a few products or producers dominate consumer engagement, often exacerbated by algorithmic promotion and user inertia (Knox & Datta, 2022; Rietveld & Schilling, 2020).

At the same time, digital platforms enable the coexistence of both superstar and niche content by reducing search costs and expanding the variety of accessible music (Brynjolfsson et al., 2010; Datta et al., 2018). This more democratized access supports not only a handful of widely followed “superstar” playlists but also a much broader set of moderately followed (“mid-tail”) and niche (“long-tail”) playlists. The mid-tail refers to playlists that have a moderate but still notable follower base, while the long-tail consists of the vast majority of playlists with small or specialized

audiences. These less prominent playlists can gain visibility as users explore new genres, moods, or activities.

It is possible that the COVID-19 pandemic shifted the distribution of attention across playlists by changing users’ listening habits. As commuting and social routines were disrupted, listeners may have changed how and when they used streaming services. On one hand, these changes could have reinforced existing habits, with users relying more heavily on familiar and prominently featured playlists, leading to an amplification of the superstar effect. On the other hand, increased time at home may have encouraged more exploration, leading users to seek out new music and niche content, potentially boosting engagement with mid- and long-tail playlists.

These competing dynamics introduce ambiguity that can be resolved through empirical testing of playlist follower changes across popularity tiers. Did users concentrate on popular playlists, or did they diversify their listening habits? This leads to our second research question: *How did the pandemic impact playlists across different levels of popularity?*

2.3 Curator Type and Follower Resilience

We next examine whether playlists curated by different types of stakeholders including platforms, major labels, independent curators, artists, and users, showed different levels of resilience in retaining or growing followers during the pandemic.

Streaming platforms like Spotify play a central role in shaping how users discover and engage with media. Leveraging extensive user data, these platforms can algorithmically curate playlists that match listener preferences, reinforcing engagement across both popular and niche segments (Datta et al., 2018; Datta & Knox, 2022). Through selective promotion on their user interfaces, such as highlighting playlists on the homepage, genre tabs, or personalized recommendations, platforms can direct user attention to specific content (Panico & Cennamo, 2022; Rietveld et al., 2019). This ability to engineer discovery may give platform-curated playlists a unique advantage in weathering adverse situations such as the COVID-19 pandemic.

Prior studies suggest that users tend to favor playlists curated by the platform itself, likely due to their visibility and personalized appeal (Aguilar et al., 2021; Pachali & Datta, 2024). These advantages may have helped platform-curated playlists retain or even grow their follower base during a time of uncertain routines and potential changing media habits. Even though other curators including major labels, professional tastemakers, artists, and everyday users, contribute to playlist diversity, they differ significantly in their ability to promote content. Major record labels, for example, can use both platform tools and external marketing to boost engagement (Boughanmi & Ansari, 2021), while smaller curators typically rely on organic discovery. Artist- and user-generated playlists may be even more constrained, often lacking algorithmic visibility or institutional support. These differences suggest that curator type could be an important determinant of how well playlists sustained engagement during the pandemic.

In this study, we define resilience as a playlist’s ability to maintain or increase its follower count in the face of pandemic-related disruptions. Given the varying resources and strategic capacities of curators, it remains an open question whether some types were systematically more resilient than others. This leads to our third research question: *How did playlists curated by music streaming platforms, major labels, artists, professional curators, independent curators, and users differ in their resilience to changes in follower numbers during the pandemic?*

2.4 Content Composition and Major Label Influence

In addition to who curates a playlist, what content it features may also shape follower shifts during the COVID-19 pandemic.

A playlist’s composition, particularly its share of tracks from major versus independent labels, can affect how appealing it is to users. Major labels, such as Universal Music Group, Sony Music Entertainment, and Warner Music Group, have long dominated the recorded music industry and continue to provide much of the most visible content on streaming platforms. Tracks from these labels benefit from brand recognition, high production quality, and familiarity, which can influence listener behavior even when users are not aware of the curator. Prior research suggests that listeners treat major label songs as “search tracks”, being reliable and easily evaluated options that help reduce uncertainty in music selection (Mu et al., 2018).

During the pandemic, when many users faced greater uncertainty and changing routines, playlists featuring a higher proportion of major label content may have gained an advantage. With limited access to live performances and fewer alternative channels for discovery, listeners might have gravitated toward playlists that featured recognizable, high-profile content. Playlists with a greater share of major label tracks, regardless of who curated them, may have offered a sense of reliability or comfort in an otherwise uncertain media environment. In contrast, playlists featuring primarily independent content may have faced greater challenges in attracting or retaining followers. This study empirically evaluates whether playlists with greater major-label representation actually retained or attracted more followers, providing evidence on the effectiveness of content branding under pandemic conditions.

This distinction between branded content and lesser-known material offers an insight into how playlist composition itself, apart from its visibility or curator, influences user engagement. We therefore ask: *Did playlists featuring a higher proportion of tracks from major labels experience an increase in followers during the pandemic?*

3 Data

To examine how the COVID-19 pandemic affected playlist engagement on music streaming platforms, we used a dataset collected from Spotify, the industry’s leading service. We rely on weekly

changes in playlist followers to capture how user behavior evolved during the pandemic and how that behavior may have influenced the visibility and positioning of different stakeholders within the platform. This section introduces the platform context, describes our dataset, and explains how our variables are operationalized to empirically answer the study’s four research questions: overall changes in follower engagement (RQ1), variation across playlist popularity (RQ2), resilience across curator types (RQ3), and the influence of major label content (RQ4).

3.1 Background on Music Streaming Platforms

The music industry is dominated by digital streaming services, which accounted for 83 percent of total revenue in the US (RIAA, 2021) and 62.1 percent worldwide (IFPI, 2021) by the end of 2020. These streaming services allow users to access millions of tracks and playlists that can be produced and curated by anyone on the platform. They have been a major form of music consumption in the US since the mid-2010s. Among these platforms, Spotify is the global market leader, with a total of 381 million global users and 172 million subscribers (MIDIa Research, 2022).

Users can browse tracks, albums, and playlists through an intuitive interface. Playlists, which are lists of tracks curated by various entities, serve as key tools for organizing content around particular themes, genres, or moods, allowing users to discover and continually play curated selections of music (Pachali & Datta, 2024; Aguiar & Waldfogel, 2021). For instance, Spotify’s ”Rock Classics” playlist offers a selection of popular rock tracks curated by the platform itself. Spotify reports that playlists account for approximately 30% of its users’ listening time (Pachali & Datta, 2024), highlighting their importance for user experience and platform engagement.

Users can “follow” playlists, enabling them to access these playlists easily in their libraries and stay updated with any changes. The following behavior not only reflects user interest and engagement, but also signals playlist visibility and influence. Playlist follower growth is driven in part by visibility and accessibility, which can result from either organic user search or platform-driven promotion. Users may find playlists by actively searching for specific themes, genres, or activities. At the same time, streaming platforms can amplify visibility by featuring selected playlists in prominent locations such as the homepage, genre tabs, or search interface. These featured placements can be organized around moods (e.g., “chill”), activities (e.g., “workout”), or genres (e.g., “pop”) to help steer user attention and engagement toward selected content.

3.2 Sample

In this study, we analyze a sample of global playlists available on Spotify from 29 October 2019 to 4 October 2020. It covers 19 weeks before and 30 weeks after the World Health Organization’s pandemic declaration on March 11, 2020.

In line with Pachali and Datta (2024), we rely on two primary datasets to assemble our sample:

(1) playlist-level follower data from *Chartmetric.com*, a third-party service that aggregates daily data from the Spotify Web API, which we aggregate into weekly intervals to observe follower changes over time, and (2) playlist feature data from *Everynoise.com/worldbrowser.cgi*, providing weekly snapshots of playlists featured on Spotify’s Search Page to control for platform-driven promotion.

The *Chartmetric* dataset provides comprehensive follower data across a wide selection of Spotify playlists. Chartmetric collects information on playlist followers, track inclusions, and curator types, covering over 1.2 million playlists on Spotify as of April 2020. We use this dataset to track changes in playlist followers over time, selecting a subset of 151,039 active playlists to ensure broad representation of Spotify’s ecosystem. This subset covers more than 90% of total followers on Spotify, providing a robust basis for analyzing changes of playlist followers over times.

To focus on consistently active playlists and ensure data continuity, we refine the sample further, selecting 104,835 playlists with 49 consecutive weeks of follower data. From this, we retain 39,918 playlists with complete metadata, with varying relevant attributes available. They include genre, curator type, and share of tracks from MLs and other playlist characteristics. This allows us to focus on playlists with sufficient information for analysis of changes in playlist followers, capturing a variety of playlist sizes, curators, and track compositions.

The *Everynoise* dataset complements this follower data by capturing weekly instances of playlists featured on Spotify’s Search Page. Featuring on the Search Page is platform-driven promotion increasing a playlist’s visibility and accessibility to users. We use this data to identify which playlists received this additional visibility, controlling for the effects of featuring when analyzing changes in playlist followers. By integrating these weekly featuring indicators, we can better isolate follower changes driven by organic user engagement from those influenced by platform-driven promotion.

While our study follows Pachali and Datta (2024) in leveraging data from *Chartmetric.com* and *Everynoise.com*, it extends beyond their sample by focusing on a broader set of playlist characteristics and covering a longer period. Specifically, Pachali and Datta’s dataset primarily centers on playlists active until early 2020, with a focus on general drivers of playlist demand and popularity across Spotify. In contrast, our sample spans from October 2019 to October 2020, capturing the weeks before and several months after the COVID-19 pandemic declaration, allowing us to specifically analyze pandemic-related shifts in playlist follower.

Although Pachali and Datta’s sample includes playlists curated by platforms and major labels, our study specifically examines the role of different curators, incorporating playlists curated by independent labels, individual artists, professional curators, and users. This broader focus enables us to explore potential differences in pandemic impacts across curator types. Furthermore, while Pachali and Datta emphasize individual tracks by major label artists and popular songs, our study analyzes playlist-level composition, examining the proportion of major label versus independent tracks within playlists. This approach allows us to assess whether content composition

within playlists influenced follower dynamics, offering new insights into content-driven engagement beyond Pachali and Datta’s focus on song-level popularity.

We report summary statistics of our playlist sample by curators in Table 2. Based on our sample, most popular playlists are Spotify-curated. Major labels (MLs) as curators do not gain any disproportionate market share of followers in the playlist market compared to other curators. This is consistent with Pachali & Datta (2024) showing that playlist followers on Spotify are highly concentrated. This concentration supports our popularity investigation and reinforces the need to distinguish between top-tier “superstar” playlists and the broader long tail apart from the curator types. Our data also reveal substantial variation in playlist composition. Despite significant variations in the proportion of major label tracks across playlists over time in our study, on average, more than half of the tracks in playlists are from major labels, except in playlists curated by independent labels. Moreover, we observe that most of the playlists featured on Spotify’s Search Page during the study period are Spotify-curated. To better understand whether these playlists attract followers organically, we include featuring as a control variable. This allows us to distinguish between follower growth driven by platform promotion and growth that may reflect underlying user interest or curator strength.

Table 2: Summary statistics of playlist sample curated by Spotify, major labels (MLs), artists, independent labels, other and users

Curators	Spotify	Pro cu- rators	MLs	Artists	Indie Labels	Other	Users
# Playlists	3,504	4,834	2,195	5,626	1,399	18,032	4,328
Avg. Followers	217,546	14,400	10,511	10,085	8,291	4,443	569
Avg. Δ Followers	669	16	28	23	54	5	0.5
Number of playlist across different popularity (i.e., share of follower percentile th)							
# in $Pr \leq 0.1$	608	18	1	16	2	0	0
# in $0.1 < Pr \leq 1$	1,581	665	220	433	92	574	0
# in $1 < Pr \leq 5$	856	1,678	880	2,185	524	8,412	1
# in $5 < Pr \leq 10$	263	1,370	643	1,897	483	6,195	2,131
# in $Pr > 10$	196	1,103	451	1,095	298	2,851	2,196
Playlist attributes and features							
Avg. MLs Share	51.46	53.74	77.81	50.78	28.31	54.82	56.55
SD. MLs Share	29.71	27.29	17.31	31.91	29.60	26.93	26.21
Avg.TotalFeature	63.21	0.43	0.88	0.04	0.02	0.01	0.00

Notes: Reported across selected 39,918 playlists in our sample. The table reports the number of playlists curated by different curators, the average (Avg.) and its weekly change (Δ) in followers, the number of playlists across different popularity, measured in percentile ranks, the average and standard deviation (SD.) of major labels’ tracks share (in %) across playlists over the study period and average number of times a playlist got featured on Spotify’s Search Page.

3.3 Variable Operationalization

Our outcome variable is the weekly change in playlist followers, as this measure captures shifts in playlist engagement within the platform over time, allowing for a temporal view of how followers responded to pandemic-related changes. By examining weekly changes rather than static follower counts, we can observe engagement patterns that reveal short-term fluctuations in user interest, which might be overlooked when relying only on a single-point snapshot. A static count might capture an overall change but could miss significant shifts such as sudden drops or surges in followers. Such changes reflect real-time user responses to specific events or conditions during the pandemic. Looking at the change also helps address behavioral inertia in digital subscriptions when users often do not actively unsubscribe or unfollow due to frictions or low incentives to change.

We define several variables that capture the specific characteristics of playlists and help us answer the research questions. First, to assess the overall shift in engagement (RQ1), we incorporate COVID-19 indicators including a binary indicator for the official pandemic declaration date, a country-weighted government stringency index, and a country-weighted residential mobility index. These indicators capture the average shift in playlist engagement following the initial pandemic declaration and the evolving impact of government restrictions and public mobility patterns. Second, to explore whether playlists with varying levels of popularity experienced different follower shifts (RQ2), we classify playlists by popularity tiers based on their follower counts. Third, to examine differences in resilience across stakeholders (RQ3), we categorize playlists by curator types. These types include Spotify, major labels, independent labels, artists, professional curators, and users. Each type represents varying levels of resources and potential influence on the platform. Fourth, to test whether content composition affected engagement (RQ4), we include the share of major label tracks within each playlist, reflecting whether well-branded content helped retain or attract followers. Finally, we control for potential confounding factors such as playlist themes, platform-based promotion (e.g., Search Page features), and seasonal effects, which allow us to isolate the impact of the pandemic from other influencing factors. Table 3 provides details about the operationalization of the variables.

3.3.1 COVID-19

To capture the impact of COVID-19 on playlist followers, we employ a combination of pandemic indicators. Using a single date, such as the World Health Organization’s pandemic declaration on 11 March 2020 ($CovidTime_t$), provides a clear start point for analyzing the initial effects. However, this date alone may oversimplify the pandemic’s effects, especially when the evolving landscape of government restrictions and public mobility patterns could influence music consumption patterns over time. A single date indicator may miss the ongoing and variable impact of lockdowns, which altered mobility and user engagement behaviors as the pandemic progressed.

To address this, we include two additional measures that account for the variation in pandemic-related restrictions experienced by users worldwide. Building on the approach used by Sim et al. (2022), we calculated a country-weighted average government restriction index ($StringencyIndex_t$) and a country-weighted average percentage change in time spent at home ($Residential_t$). The Government Response Stringency Index from the Oxford COVID-19 Government Response Tracker quantifies government restrictions by factoring in measures such as school closures, workplace closures, and travel bans (Wenham et al., 2023), thus offering a more granular view of restriction intensity and timing across countries. The Google COVID-19 Community Mobility Report’s *Mobility trends for places of residence* data ($Residential_t$) complements this by measuring changes in the amount of time individuals spent at home, reflecting users’ exposure to and engagement with digital media during prolonged stay-at-home orders (Google, 2022).

Given that our data contains only playlists available to a global audience, constructing country-weighted measures for both $StringencyIndex_t$ and $Residential_t$ allows us to reflect the influence of restrictions based on the relative streaming activity from each country. A simple unweighted average would not capture the variations in user engagement across different regions; thus, we apply country weights to emphasize the proportional impact of each country’s restrictions according to its streaming volume. This approach gives greater weight to countries with higher streaming activity, aligning our metrics with the international scope of our playlist sample.

We calculate the country-weighted government restriction and residential mobility indices as follows:

$$StringencyIndex_t = \sum w_{c,t} * StringencyIndex_{c,t}$$

$$Residential_t = \sum w_{c,t} * Residence_{c,t}$$

where $w_{c,t}$ is the proportion of total streams of the top 200 streamed songs in country c in period t relative to the total top 200 streamed songs across all available countries. This country-weighted structure enables us to capture the dynamic and regionalized effects of the pandemic on playlist followers, reflecting how varying restrictions influenced playlist engagement within the platform globally.

Integrating these three indicators, pandemic declaration date, government restriction levels, and residential mobility changes, allows us to estimate the varying impacts of COVID-19 on playlist followers. The variations of these factors enable us to analyze not only the overall average effects of the pandemic but also the influence of ongoing restrictions and behavior shifts on streaming platform engagement.

3.3.2 Playlist Popularity

To assess the impact of the COVID-19 pandemic on playlists with varying levels of popularity, we categorize playlist popularity according to follower counts, capturing different levels of prominence within the music streaming platform. This approach enables us to distinguish between highly

popular ‘superstar’ playlists, moderately popular ‘mid-tail’ playlists, and ‘long-tail’ playlists with smaller follower counts. These categories reflect differing levels of initial popularity and visibility across Spotify.

Following Elberse and Oberholzer-Gee (2006), Brynjolfsson et al. (2011), and Kumar et al. (2014), we utilize percentile ranks of follower counts to capture these dynamics. Based on a comprehensive list of approximately 1.2 million playlists tracked in April 2020, we define five popularity groups as follows:

Top 0.1% – “Superstar” playlists. These account for over 50% of total followers despite representing a tiny share of all playlists.

0.1% to 1% – Highly popular playlists just below the superstar tier. Together, the top 1% of playlists account for roughly 80% of all followers.

1% to 5% – “Mid-tail” playlists with moderate followings. Cumulatively, the top 5% of playlists account for about 97% of all followers.

5% to 10% – Lower mid-tier playlists with more limited reach. The top 10% of playlists together account for approximately 99% of followers.

Bottom 90% – “Long-tail” playlists with niche or small audiences. Collectively, they represent the remaining 1% of total followers.

This categorization reflects the highly skewed follower distribution typical of digital platforms, where playlist follower’s distribution is concentrated among a few top entities. Using data from April 2020 as a baseline for categorization is appropriate because it reflects follower distribution shortly before the pandemic’s widespread impact on global routines and media consumption patterns. This timing allows us to capture playlists’ established popularity without significant pandemic-driven influences. By excluding playlists introduced after April 2020, we ensure that our sample includes only those playlists active prior to the pandemic, creating a stable reference for analyzing changes in followers. This approach enables us to assess whether the pandemic intensified the superstar effect, with popular playlists gaining even more traction, or contributed to a more balanced engagement across niche, long-tail playlists.

3.3.3 Curators

To address our research question on playlist curators and the pandemic’s impact on playlist followers, we categorize playlists based on curator types. This enables us to investigate whether playlists curated by different stakeholders demonstrated varying resilience or experienced distinct advantages during the pandemic. Specifically, our study examines whether well-resourced curators like Spotify and major labels retained or increased their follower growth more effectively than less-resourced groups, such as independent labels and user-generated playlists.

Pachali and Datta (2024) provide a classification of curators that we directly adopt to differentiate playlists by curator type in our analysis. While their study uses this classification to explore

general playlist popularity and demand, we place curator type at the center of our analysis to examine differential impacts of the pandemic on follower engagement across these groups. Using their categorization, we classify curators as follows:

Spotify: Playlists curated directly by Spotify, often by in-house teams specializing in global music trends, account for much of the platform’s popular content. These playlists generally feature a significant proportion of tracks by the label’s artists, which may enhance visibility and engagement.

Major Labels: Major record labels, including Universal Music Group, Sony Music Entertainment, and Warner Music Group, curate playlists under label-specific brands such as “Digster” for Universal, “Filtr” for Sony, and “Topsify” for Warner. These playlists typically incorporate a higher proportion of tracks from the respective labels’ artists, which may contribute to their visibility and follower retention within the platform.

Independent Labels and Artists: Independent labels and individual artists curate playlists under their own names or smaller label brands.

Professional Curators: Professional curators, sometimes known as “tastemakers,” include brands, media companies, or radio stations that create playlists to complement their brand or broadcast content. As defined by Pachali and Datta (2024), tastemakers have a reach of at least 50,000 followers, indicating a notable level of platform influence.

Users and Other Curators: User-created playlists, typically designed for personal music organization, have a reach of 1,000 followers or fewer. The “Other Curators” category includes playlists that do not fit the previous classifications, covering a diverse range of less common playlist types.

By examining changes in playlist followers across these curator types, we assess whether curators with greater algorithmic visibility, promotional resources, or brand recognition were better able to retain or grow their audiences during the pandemic, providing empirical insights on stakeholder resilience and competitive dynamics within the platform ecosystem.

3.3.4 Major Labels Share

The global music industry is significantly shaped by the influence of three major recorded labels: Universal Music Group, Sony Music Entertainment, and Warner Music Group. Together, these companies held more than 68 percent of the total recorded music market share worldwide in 2020 (Music & Copyright, 2021), demonstrating their dominance in the industry. To explore the impact of major label presence within playlists on follower change during the pandemic, we measure the proportion of major label tracks within each playlist rather than counting individual tracks or focusing on specific, high-profile artists.

This approach offers a few key advantages. First, by focusing on the share of tracks from major labels rather than the number of songs or specific artists, we establish a consistent metric that allows for comparison across playlists of various sizes and compositions. This metric better

captures the general influence of major labels across playlists without conflating results with individual superstar effects. Although previous research, such as Pachali and Datta (2024), has found that superstar artists from major labels can influence playlist popularity, our study is concerned with the broader impact of major label presence on playlist follower growth. By examining the aggregated share of major label tracks, we aim to understand whether playlists with a greater proportion of content from well-established labels demonstrated resilience or increased appeal during the pandemic period.

The long-standing market presence of these three major labels, with roots extending back to the 1970s (Smith & Telang, 2016), suggests that their inclusion within playlists serves as a quality signal. Playlists featuring a higher share of major label tracks are likely perceived as containing reliably curated, popular, or high-quality content, benefiting from these labels' established brand reputations and promotional reach. This signal of quality could make such playlists more appealing to users, especially in uncertain times when familiar content might be preferred. This aligns with our research question on whether major label tracks could potentially attract more followers during the pandemic, shedding light on how content composition within playlists influenced engagement levels in response to changes in listening behaviors.

3.3.5 Control Variables

To ensure an accurate analysis of how COVID-19 impacted playlist followers, we include several control variables that could otherwise influence follower dynamics independently of our primary variables of interest. By controlling for these factors, we aim to isolate the effects of playlist popularity, curator type, and major label track share, which are central to our study's research questions.

First, we control for playlist categorization, as playlists associated with different genres, moods, and activities likely attract distinct listener demographics and engagement levels. Following the categorization method established by Pachali & Datta (2024), we assign each playlist to one or more of 32 categories observable on Spotify's Search Page, which plays an integral role in Spotify's discovery and curation processes. These categories include genre-based themes (e.g., "Afro," "Jazz"), mood-based themes (e.g., "Chill," "Romance"), and activity-based themes (e.g., "Workout," "Dinner"), all significant to how users navigate playlists on the platform. Our selected categories span a broad range of musical themes and use cases, such as: (1) Activity, (2) Afro, (3) Anime, (4) Arab, (5) Blues, (6) Chill, (7) Classical, (8) Country, (9) Dance, (10) Desi, (11) Dinner, (12) EDM, (13) Focus, (14) Funk, (15) Gaming, (16) Hip Hop, (17) Indie, (18) Inspirational, (19) Jazz, (20) K-pop, (21) Latin, (22) Metal, (23) Mood, (24) Party, (25) Pop, (26) Punk, (27) R&B, (28) Rock, (29) Romance, (30) Sleep, (31) Soul, and (32) Workout.

The integration of these thematic categories reflects the diverse interests and consumption patterns of Spotify's users. For instance, genre-based playlists such as "Pop" might attract a broader

audience and maintain greater follower stability, while activity-based playlists like “Workout” may see more fluctuations in engagement, especially as user routines shifted during the pandemic. By controlling for these thematic categories, we aim to reduce potential confounding effects, ensuring our analysis accurately isolates the impacts of curator type, playlist popularity, and major label composition on follower dynamics.

Second, we control for the effect of being featured on Spotify’s Search Page. Playlists featured prominently on the platform receive enhanced visibility, which can drive significant gains in follower counts (Pachali & Datta, 2024; Aguiar & Waldfogel, 2021). This feature effect could amplify follower engagement regardless of a playlist’s popularity or curator type, potentially obscuring the specific impact of pandemic-induced shifts. Including this control helps clarify whether any observed resilience or growth in followers is due to feature-driven exposure rather than intrinsic characteristics of the playlist or its curator.

Third, we control for seasonal and time-based trends by including variables for the months of the year and specific weeks within our study period. These controls account for general growth trends in platform usage as well as any seasonal variations that could affect streaming activity across the year. By incorporating these temporal factors, we aim to isolate the impact of the pandemic on follower growth, reducing the potential influence of natural growth patterns and seasonality on our analysis.

Together, these control variables allow us to isolate the effects of playlist composition, curator identity, and popularity levels, providing a robust framework for assessing how the pandemic influenced follower dynamics across different stakeholder groups and playlist types.

Table 3: Variable Operationalization

Variable	Operationalization
Dependent Variable	
$\Delta \text{Followers}_{i,t}$	A weekly change in number of follower of playlist i in period t compare to period t-1.
Independent Variables	
CovidTime_t	Step dummy variable indicating the beginning of the pandemic, equaling 1 if period t is after 11 March 2020, 0 otherwise.
StringencyIndex_t	Country-weighted average government restriction in period t rescaled to a value from 0 to 100 (100 is strictest response). $\text{StringencyIndex}_t = \sum w_{c,t} * \text{StringencyIndex}_{c,t}$ where $w_{c,t}$ is the proportion of total streams of top 200 streamed song in country c in period t relative to total top 200 streamed songs across all recorded countries
Residential_t	Country-weighted average percentage change of time spent at home from the baseline in period t. $\text{Residential}_t = \sum w_{c,t} * \text{Residence}_{c,t}$ where $w_{c,t}$ is the proportion of total streams of top 200 streamed song in country c in period t relative to total top 200 streamed songs across all recorded countries
Popularity_i	Categorical variable indicating the popularity of playlist i. Playlist i either be (1) Playlists from first top 0.1% covering 52% of total followers (2) Playlists from the the top 0.1% to the top 1% covering 84% of total followers (3) Playlists from the top 1% to the top 5% covering 97% of total followers (4) Playlists from the top 5% to the top 10% covering 99% of total followers and (5) Playlists that are not included in top 10%
Curator_i	Categorical variable indicating the curator of playlist i. Playlist i either be curated by (1) Spotify, (2) the major recorded labels (MLs), (3) artists, (4) independent labels, (5) professional curators, (6) users and (7) others
$\text{MLShare}_{i,t}$	The percentage of tracks produced by Universal Music Group, Sony Music Entertainment, or Warner Music Group for playlist i in period t
Control Variables	
$\text{Categorization}_{ik}$	Categories of playlist i operationalized as grand-mean centered variable (%) for category k in all selected 32 categories. The measure implies the difference from the average proportion of all playlists that belongs to a certain category or the degree of being able to substitute to a particular genre.
$\text{Featured}_{i,t}$	On average, how many times playlist i got featured in a session on Spotify’s Search Page across countries in period t.
Month_t	months of the year in period t.
Time_t	Time trend for period t.

4 Empirical Analysis and Results

4.1 Identification Strategy

Although the pandemic represents an exogenous shock to Spotify’s ecosystem, identifying its impact requires addressing several challenges. First, because the pandemic affected all playlists at the same time, we lack a conventional control group with untreated units for a difference-in-differences analysis. Second, because our sample is limited to global playlists, we miss the potential for observing variations in engagement that country-level playlists might reveal. With country-specific playlists, we could observe how regional differences in government responses and mobility restrictions at a given time directly influenced playlist followers, offering a clearer picture of localized pandemic effects.

To address these challenges, we employ a unit fixed effects model on playlists, which effectively controls for time-invariant playlist-specific characteristics that may influence follower dynamics (Papies et al., 2022). This approach is appropriate for our analysis due to the following reasons:

First, by estimating the model with playlist (i.e., unit) fixed effects, identification relies on within-playlist variation over time. This means that each playlist’s pre-pandemic follower trajectory functions as its own counterfactual benchmark, eliminating bias from time-invariant unobserved heterogeneity. The lack of a separate untreated group is thus mitigated by using historical patterns of the same unit as the basis for comparison. Second, the model accounts for unique, unobserved attributes of each playlist, such as its danceability attributes or curation styles, which could independently affect follower engagement regardless of pandemic influences. By controlling for these time-invariant characteristics, our model isolates changes in follower counts that are attributable to pandemic-related shifts. Third, the model includes weekly time controls, which capture platform-wide trends and other temporal fluctuations that may impact playlist engagement. This enables us to disentangle the pandemic’s impact on playlist followers from growth trends, seasonal variations, and other time-based influences. Identification therefore comes from deviations in follower growth within the same playlist during the pandemic relative to its own pre-pandemic baseline, net of seasonal and platform-wide fluctuations.

While our data are aggregated at the playlist level and do not include direct country-level identifiers, we account for international variation in pandemic exposure by constructing country-weighted indices of government stringency and residential mobility. These indices reflect the relative streaming volume from each country, giving more weight to countries with higher user activity. This approach allows us to indirectly capture the influence of country-specific restrictions and behavioral shifts on global playlists, aligning the pandemic measures with the geographic distribution of actual engagement. Although we cannot observe individual country-level playlist trends, this weighting strategy ensures that our analysis incorporates meaningful cross-country variation in pandemic-related disruptions.

In summary, our identification strategy combines a unit fixed effects model, weekly time controls, and carefully defined control variables (including genre categories) to capture unique, time-invariant characteristics of each playlist while isolating the specific impacts of the pandemic on playlist followers.

4.2 Empirical Model

To assess the impact of the COVID-19 pandemic on playlist followers and examine how different stakeholders, including curators, content producers, and playlists with different levels of popularity, were affected, we employ a log-log model with (unit) fixed effects. This specification allows us to interpret the estimated coefficients as elasticities (Papies & van Heerde, 2017), capturing the relative changes in playlist followers in response to pandemic-related factors. The log transformation also helps correct for the skewed distribution of follower counts, particularly across playlists of vastly different popularity tiers. Our model is designed to address the four research questions by isolating within-playlist changes while controlling for time-invariant characteristics and time-specific effects. It also enables us to test for differential effects across playlists grouped by (percentile-based) popularity, curator type, and content composition. The model is specified as follows:

$$\begin{aligned}
\Delta \ln(Follower_{i,t}) = & \kappa Covid_t \\
& + \sum_{j=1}^4 \gamma_j Covid_t \times Popularity_i \\
& + \sum_{k=1}^6 \beta_k Covid_t \times Curator_i \\
& + \eta_1 MLShare_{i,t} + \eta_2 Covid_t \times MLShare_{i,t} \\
& + \sum_{l=1}^{32} \alpha_l Covid_t \times Categorization_{ik} + \alpha_{33} \ln Featured_{i,t} \\
& + \alpha_{34} Month_t + \alpha_{35} \ln Time_t + \varepsilon_{i,t}
\end{aligned}$$

In this specification, $Covid_t$ represents the pandemic-related variable, defined in three indicators: $CovidTime_t$ (pandemic declaration date), $StringencyIndex_t$ (weighted government restriction index), and $Residential_t$ (weighted time spent at home). Each measure of $Covid_t$ enables us to explore how different aspects of the pandemic affected follower engagement. Specifically, κ reflects the fixed effect of the pandemic declaration, government restrictions, and changes in home-based activity on changes in followers.

The interaction terms $Covid_t \times Popularity_i$ capture the pandemic's differential impact across

playlists with varying levels of popularity, allowing us to assess whether highly followed “super-star” playlists (top 0.1%), moderately followed “mid-tail” playlists, or niche “long-tail” playlists exhibited different follower trends. Similarly, $Covid_t \times Curator_i$ measures pandemic effects across curator types, capturing whether playlists curated by Spotify, major labels, or other stakeholders showed varying resilience or follower growth in response to the pandemic.

We include the variable $MLShare_{i,t}$ to measure the influence of major label content in playlists. The interaction $Covid_t \times MLShare_{i,t}$ allows us to assess whether playlists with a higher share of major label tracks were particularly resilient or attractive to followers during the pandemic, especially given the perceived quality associated with major label content.

The $Covid_t \times Categorization_{ik}$ interaction terms control for playlist categorization (e.g., genre, mood, or activity), which ensures our analysis focuses on changes in playlist followers specific to curator type, popularity, or major label composition rather than thematic variations. Additionally, the $\ln(Featured_{i,t})$ variable controls for any playlist featured on Spotify’s Search Page, which often results in significant follower gains. By isolating these effects, we can distinguish feature-driven gains from organic user engagement during the pandemic.

The model also incorporates monthly and weekly controls to account for seasonal trends and overall platform growth over time. The inclusion of these controls helps ensure that our estimates reflect the pandemic’s specific impact on follower engagement, adjusting for temporal trends in playlist engagement on streaming platform.

Given the absence of an untreated control group and the global nature of Spotify’s audience, we use a unit fixed effects model to control for unobserved playlist-specific characteristics that may influence follower trends independently of the pandemic. This approach stabilizes any time-invariant, and playlist-specific factors in order to ensure that our results capture only within-playlist changes in response to pandemic-related shifts in user behavior.

4.3 Results

Table 4 shows the estimated fixed effects of the pandemic on playlist followers across different popularity and relevant platforms’ stakeholders, including curators and track producers. Using a unit fixed effects model, we analyze how the pandemic declaration, global government restrictions, and changes in time spent at home globally influenced playlists across different curator types and popularity levels. To facilitate interpretation, we reported total estimated effects for playlists by curator and popularity category, as illustrated in Figure 1 - 3.

Columns (1), (3), and (5) in Table 4 indicate the average pandemic effects (i.e., pandemic declaration, global government restrictions and global changes in duration spent in places of residence) on playlist followers addressing our first research question on changes in playlist followers on the streaming platform. Meanwhile, Columns (2), (4), and (6) explore effects across different platform stakeholders, examining variations by curator type, playlist popularity, and major label

track share.

It is worth noting that while the *MLShare* variable shows statistical significance in some cases, its effect size remains relatively small. Specifically, a 1% (or 0.01 unit) increase in the proportion of major label tracks within a playlist yields only modest shifts in follower growth, suggesting that this measure captures subtle adjustments rather than substantial changes in follower dynamics. We maintain this scale for *MLShare* to ensure consistency across the model and facilitate direct comparisons with other variables.

In the following sections, we further discuss these results, first focusing on overall follower changes, followed by effects across popularity levels and curator types, and concluding with the impact of major label track share on follower growth. Since playlist popularity and curator type are often interdependent meaning that popular playlists are frequently curated by specific types of stakeholders (e.g., Spotify itself or major labels), we analyze these categories jointly rather than separately to more accurately reflect the combined influence of both factors on changes of playlist followers during the pandemic.

4.3.1 Effects on Playlist Followers

Even though the number of music streaming subscribers worldwide grew from 2019 to 2020, on average, we do not see any significant change in playlists' followers after the pandemic declaration (Column (1) in Table 4). However, there is a significant ($p < .05$) positive relationship between an increase in playlist followers and stay-home measures, including stricter government policies and more time spent in residential places (Columns (3) and (5) in Table 4). Specifically, during the pandemic period, assuming the mean (i.e., median) global stringency index equals 65 on a certain week, each playlist gets 0.0004% increase in followers. For a mean (i.e., median) increase of 8% in time spent at home during the pandemic, each playlist get 0.0024% increase in followers.

This relationship suggests that user engagement with playlists may have been influenced more directly by restrictions on mobility and the resulting increase in home-based activities rather than the announcement of the pandemic itself. The initial declaration may not have significantly impacted listening habits, as it was only over time, as restrictions tightened and people adjusted to prolonged periods at home, that users sought greater engagement with digital content like streaming playlists. This aligns with behavioral trends observed in other digital media, where consumption patterns shifted notably as people adapted to "stay-at-home" lifestyles. Additionally, the positive relationship with time spent at home may reflect a broader shift in daily routines. With reduced commutes and increased flexibility in working or studying from home, listeners may have sought new music experiences, exploring or following more playlists.

These findings suggest that while the pandemic declaration initiated the beginning of a shift in user behavior, it was the ongoing restrictions and home-based adjustments that likely played a more substantial role in driving increased engagement with playlists (via followers) over time.

Table 4: Estimated Results from a Unit Fixed Effects Model (Within Playlists)

Model:	CovidTime (1)	CovidTime ¹ (2)	Stringency (3)	Stringency ¹ (4)	Residential (5)	Residential ¹ (6)
Covid _t ^a	0.0002 (0.0001)	-0.0007 (0.0004)	0.00000** (0.00000)	-0.00001 (0.00001)	0.00003** (0.00000)	0.00001 (0.00002)
Covid _t × Artists		-0.0019** (0.0004)		-0.00002*** (0.00001)		-0.00003 (0.00002)
Covid _t × Indie Labels		-0.0024*** (0.0005)		-0.00003*** (0.00001)		-0.00004 (0.00003)
Covid _t × ML		-0.0018*** (0.0005)		-0.00002*** (0.00001)		-0.00002 (0.00003)
Covid _t × Other		-0.0023*** (0.0004)		-0.00003*** (0.00001)		-0.00006*** (0.00002)
Covid _t × Pro Curators		-0.0031*** (0.0004)		-0.00004*** (0.00001)		-0.00005*** (0.00003)
Covid _t × Users		-0.0027*** (0.0004)		-0.00004*** (0.00001)		-0.00010*** (0.00002)
Covid _t × top 1%		0.0001 (0.0005)		0.00000 (0.00000)		-0.00001** (0.00002)
Covid _t × top 1-5%		0.0019** (0.0006)		0.00003*** (0.00001)		0.00007*** (0.00003)
Covid _t × top 5-10%		0.0027*** (0.0006)		0.00004*** (0.00001)		0.00010*** (0.00003)
Covid _t × Tail		0.0028*** (0.0006)		0.00004*** (0.00001)		0.00014*** (0.00000)
MLShare ^a		0.0000 (0.0000)		0.00002 (0.00002)		0.00003* (0.00002)
Covid _t × MLShare ^a		0.0000*** (0.0000)		0.00000*** (0.00000)		-0.00000*** (0.00000)
Genre	Y	Y	Y	Y	Y	Y
Feature	Y	Y	Y	Y	Y	Y
Month	Y	Y	Y	Y	Y	Y
Trend	Y	Y	Y	Y	Y	Y
<i>Fit statistics</i>						
Observations				1,917,624		
Adjusted R ²	0.0277	0.0279	0.0276	0.0279	0.0276	0.0278

1 Spotify is a based curator with superstar (top 0.1%) playlists.
a Coefficients reported as 0.0000 are rounded for presentation; exact estimates are smaller than 0.00001 but statistically significant.
Heteroskedasticity-robust standard-errors in parentheses
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

4.3.2 Effects on Playlist Curators across Different Popularity

We illustrated the average effects of the pandemic declaration, the effect of government restrictions (at the mean) and the effect of time spent at home (at the mean) on playlists' followers curated by different curators across different popularity in Figure 1, 2 and 3 respectively.

Estimated effects of the pandemic declaration (Figure 1) and government restrictions (Figure 2) on playlists curators and playlist popularity are consistent. Spotify did not experience any significant (i.e., inconclusive confidence interval including zero) negative impact and their playlists along the mid-tail (i.e., top 5th - top 10th of follower percentile) and long-tail gained significantly more followers, suggesting that playlists curated by the music streaming platform were resistant and favorable during the pandemic time. Specifically, on average, Spotify playlists in top 5th percentile gain a 0.16% increase in their followers and over 0.2% increase for those in top 10% and along the long-tail.

Conversely, playlists curated by major labels, professional curators, and independent curators faced more challenges in maintaining their follower base. For instance, superstar playlists (top 0.1st percentile) curated by major labels saw a decline of over 0.2% in weekly follower growth, translating to approximately 2,000 fewer followers for a playlist with a 1,000,000-follower base. Professional and independent curators appeared to suffer even more, with their popular playlists experiencing a drop in weekly follower growth of over 0.3%. These findings are consistent across both the pandemic declaration and government restriction effects, suggesting that Spotify-curated playlists, unlike those curated by other stakeholders, maintained a favorable position in attracting listeners.

Estimated effects of changes in duration spent in places of residence (Figure 3) support that Spotify playlists curated by Spotify significantly benefited from users spending more time at home. However, contrary to the impacts of the pandemic declaration and government restrictions, the time spent at home (at mean) does not have any significant effect on popular playlists. On the other hand, it tended to have a positive effect on overall less popular playlists (Playlists that are not in top 5th percentile). This shift in playlist followers patterns could suggest that listeners, with more time at home, explored beyond mainstream playlists to discover new or lesser-known content, possibly as a way to diversify their listening habits during an extended period of isolation.

From these findings, we infer that the pandemic did not reinforce the superstar effect but rather democratized playlist popularity by elevating follower growth for playlists in the mid-tail and long-tail. Spotify-curated playlists demonstrated notable resilience during the pandemic, while major labels and artists managed to alleviate some negative impacts more effectively than other curators, including professional and independent curators. This divergence may reflect the inherent advantage of being a well-resourced, platform-based curator, such as Spotify, which curates its own playlists and controls discovery algorithms. Unlike other curators, Spotify playlists might have benefited from the platform's recommendation algorithms and search visibility, which could have

amplified their exposure and follower retention during a period when users were seeking more stable, reliable content.

These results suggest that, while major labels and artists showed adaptability or relatively less negative effects on their followers' growth, Spotify's playlists were outstandingly positioned to gain higher growth in user engagement during the pandemic. In other words, the pandemic reinforced the platform's resilience and influence within its own ecosystem.

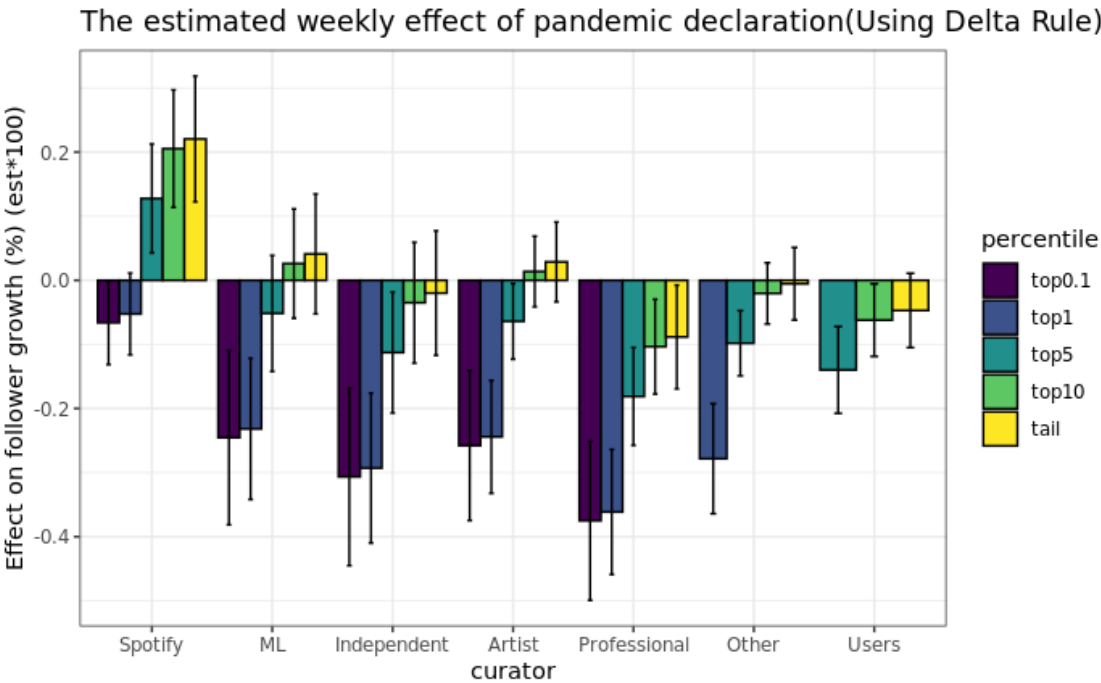


Figure 1: The estimated weekly effect of pandemic declaration

Notes: Total estimated effects from pandemic declaration by curator and popularity category, derived from the unit fixed-effects regression model. Standard errors are computed using the delta method by applying a first-order Taylor series approximation to obtain the variance of nonlinear combinations of estimated parameters.

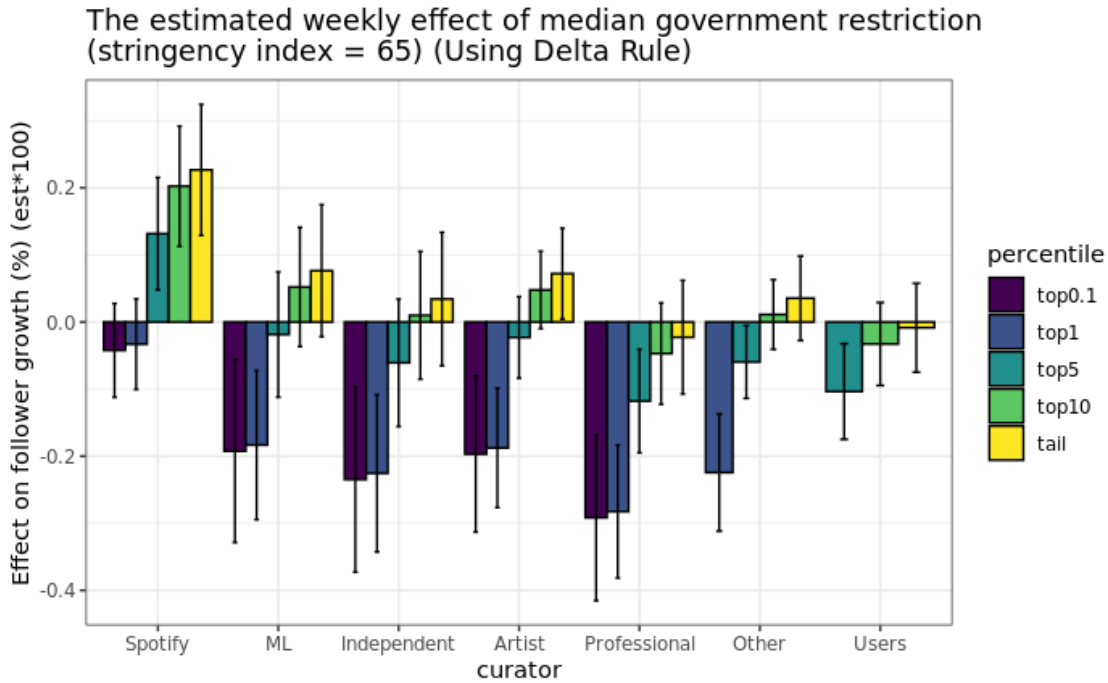


Figure 2: The estimated weekly effect of median government restriction
Notes: Total estimated effects from government restriction (with median of stringency index equivalent to 65) by curator and popularity category, derived from the unit fixed-effects regression model. Standard errors are computed using the delta method by applying a first-order Taylor series approximation to obtain the variance of nonlinear combinations of estimated parameters.

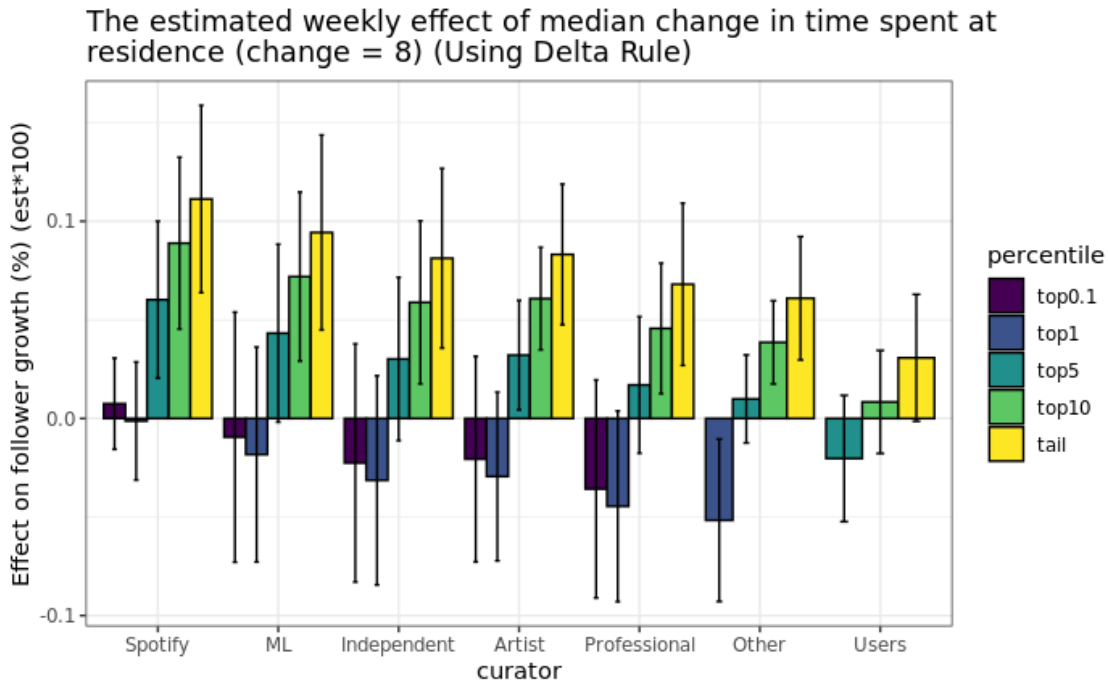


Figure 3: The estimated weekly effect of median change in time spent at home
Notes: Total estimated effects from time spent more at home (with median equivalent to 8) by curator and popularity category, derived from the unit fixed-effects regression model. Standard errors are computed using the delta method by applying a first-order Taylor series approximation to obtain the variance of nonlinear combinations of estimated parameters.

4.3.3 Effects on Content Producers

Our analysis explores whether playlists with a higher proportion of major label tracks demonstrated increased follower engagement during the pandemic. While we initially anticipated that playlists featuring more tracks from major labels would benefit from heightened appeal, the results suggest a mixed implication. Although we observe statistically significant ($p < .05$) positive effects from both the pandemic declaration and stricter government restrictions on follower growth for playlists with a greater major label share (Columns (2) and (4) in Table 4), these effects are economically small. Each 1% increase in major label track share is associated with only a marginal rise in follower counts, indicating that the role of major labels during this period, while detectable, may not have been substantial in shaping follower dynamics.

However, we observe significant ($p < .05$) negative effects associated with increased time spent at home (Column (6) in Table 4). This could indicate that, as users increased their time at home and had greater flexibility to explore less mainstream content, the appeal of playlists heavily featuring major label tracks may have diminished marginally. Users might have shifted attention toward a broader variety of content, potentially favoring independent or niche tracks that aligned more closely with their evolving daily routines or moods during an extended period of home confinement.

To further disentangle the effects of track popularity from label affiliation, we extended our model to include *AverageTrackShare*, a variable representing the average number of playlists on which each track appears on the platform (Table 5). By incorporating this measure, we can observe the impact of track popularity independently of major label affiliation. This additional variable indicates, on average, the number of playlists to which tracks from this list were added on Spotify. It is worth noting that there is no multicollinearity issue as the correlation between *MLShare* and *AverageTrackShare* is not high (i.e., 0.48). Once *AverageTrackShare* is included, the effect of major label share becomes statistically insignificant ($p > 0.1$), while *AverageTrackShare* itself is positively and significantly associated with follower growth. This result indicates that track popularity, rather than label branding, may have played a more critical role in follower engagement during the pandemic.

These findings suggest that, although major labels typically exert substantial influence in the music industry, their dominance was not evident on Spotify during the pandemic. With greater access to a wide range of content and more discretionary time, users appear to have responded more to the popularity and discoverability of individual tracks than to label affiliation. Playlists featuring widely included (and presumably popular) tracks attracted more followers, highlighting a shift toward content-driven engagement. In this context, the pandemic may have temporarily leveled the playing field, diminishing the branding advantage of major labels and emphasizing the appeal of high-rotation tracks regardless of their source.

Table 5: Excerpt of Estimated Results from a Unit Fixed Effects Model (Within Playlists) with An Addition of Track Popularity Proxy

Model:	CovidTime ¹	Stringency ¹	Residential ¹
MLShare	-0.0000 (0.0000)	0.00000 (0.00000)	0.00000 (0.00001)
Covid _t × MLShare	0.0000 (0.0000)	-0.00000 (0.00000)	-0.00000*** (0.00000)
AvgTrackShare	0.0010* (0.0004)	0.0010** (0.0005)	0.00130** (0.00055)
Covid _t × AvgTrackShare	0.0005*** (0.0001)	0.00001*** (0.00000)	0.00002*** (0.00001)
<i>Fit statistics</i>			
Observations	1,917,624	1,917,624	1,917,624
R ²	0.0279	0.0279	0.0278
Within R ²	0.0024	0.0023	0.0022

1 Spotify is a based curator with superstar (top 0.1%) playlists.

Heteroskedasticity-robust standard errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

5 Conclusion and Discussion

This study aimed to empirically examine how the COVID-19 pandemic affected Spotify’s platform engagement by focusing on changes in playlist followers. Playlists are a key mechanism through which users discover and engage with music on streaming platforms. Changes in their follower counts can therefore reflect shifts in user behavior and the relative visibility of different stakeholders during the pandemic.

To structure our discussion, we revisit the study’s four research questions, highlighting key findings and their implications. We then outline practical recommendations for industry stakeholders and conclude by reflecting on the study’s limitations and directions for future research.

RQ 1: How did the number of followers for playlists change during the pandemic?

Although global subscriptions to music streaming platforms increased during the COVID-19 pandemic (RIAA, 2021), our results show no significant change in the average growth of playlist followers. This suggests that the rise in total platform usage did not translate uniformly into increased engagement with playlists within the platform.

The lack of overall change may reflect shifts in listening contexts. As prior research indicates, user engagement with music streaming is strongly influenced by daily routines, such as commuting or working environments (e.g., Sim et al., 2022; Denk et al., 2022). During the pandemic, while some new users may have joined the platform via home-based devices (e.g., smart TVs) (Spotify,

2021), others may have reduced their listening due to the loss of music-integrated routines (e.g., driving, gym use). This disconnect between continued subscription growth and stagnant playlist follower trends may reflect a form of behavioral inertia, a situation where users remain subscribed but reduce active engagement. This effect has been documented in other digital services as well (Rietveld & Schilling, 2020).

RQ 2: How did follower dynamics differ across playlists with varying levels of popularity during the pandemic?

Our findings reveal that the most-followed “superstar” playlists did not experience a significant increase in followers during the pandemic. Instead, playlists in the mid-tail and long-tail segments saw greater relative growth, particularly during periods when users spent more time at home. This pattern suggests that the pandemic did not amplify the existing superstar effect (Rosen, 1981; Elberse & Oberholzer-Gee, 2006); rather, it fostered broader playlist engagement across less prominent playlists, leading to a more balanced follower distribution.

This shift supports prior literature on digital content consumption, which highlights how lower discovery costs and increased time spent online can facilitate the exploration of diverse or niche content (Brynjolfsson et al., 2010). As discussed in our introduction, playlists play a central role in shaping listener behavior on streaming platforms by offering accessible, theme-based curation. During the pandemic, the extended time spent at home may have encouraged users to explore beyond highly visible content, resulting in increased engagement with mid-tail playlists, which are neither widely popular nor entirely obscure.

These findings indicate that user engagement during the pandemic became less concentrated on superstar playlists, highlighting the platform’s potential to drive attention toward a broader range of content when listening behavior becomes more exploratory.

RQ 3: How did playlists curated by music streaming platforms, major labels, and independent curators differ in their resilience to changes in follower numbers during the pandemic?

Our findings show that playlists curated by Spotify demonstrated greater resilience and even achieved modest gains in follower counts during the pandemic. In contrast, playlists curated by major labels, professional tastemakers, and independent curators experienced significant declines in weekly follower growth. This pattern is consistent with prior research (e.g., Pachali & Datta, 2024; Aguiar & Waldfogel, 2021), which highlights the advantages of platform-curated content in terms of visibility, algorithmic promotion, and alignment with user preferences.

As discussed in the introduction, Spotify plays a dual role as both platform and curator, with its access to internal promotion mechanisms such as playlist featuring. Even after controlling for whether a playlist was featured on the platform, our results show that Spotify-curated playlists were less negatively affected by the pandemic and in some cases even gained followers. In contrast, major label-curated “superstar” playlists (i.e., top 0.1% by follower count) experienced declines of over 0.2% in weekly follower growth, which was equivalent to several thousand followers for a

playlist with one million followers. Professional curators and independent labels, which typically have fewer resources and limited promotional visibility, saw even larger drops in follower engagement across popularity tiers. These findings suggest that curator types differed in their resilience to changes in follower numbers during the pandemic.

RQ 4: Did playlists featuring a higher proportion of tracks from major labels experience greater follower growth during the pandemic?

Our findings reveal that playlists with a higher share of tracks from major labels showed limited follower growth during the pandemic. While there were some statistically significant positive effects linked to government restrictions and platform activity, the effect sizes were very small, meaning that each 1% increase in major label track share corresponded to only marginal changes in weekly follower growth. This suggests that the presence of major label content alone did not strongly influence user engagement.

When we examined periods in which users spent more time at home, we observed a negative association between major label track share and follower growth. This pattern may reflect a shift in user preferences toward more diverse or independent content as users engaged more actively with the platform during extended home-based routines. As discussed in our introduction, playlists serve as a key interface between listeners and content producers, and the presence of well-known label content is often viewed as a signal of quality. However, our findings indicate that during the pandemic, this signal may have been less influential in shaping follower behavior.

Further analysis that included a proxy for track popularity, which is defined as how frequently tracks appeared across playlists, showed that this popularity measure, rather than label affiliation, better explained follower growth. Once track popularity was controlled for, the effect of major label share became statistically insignificant. This suggests that users responded more to recognizable or frequently featured tracks than to whether those tracks came from major labels.

Together, these results imply that playlists featuring a higher proportion of tracks from major labels did not experience greater follower growth during the pandemic. Instead, follower growth was more strongly associated with the inclusion of popular tracks, regardless of label affiliation.

Empirical Implications

Our findings have several implications for platform managers and content producers, particularly in navigating user engagement during periods of restricted mobility and changing routines from government restrictions.

First, the observed increase in follower growth among mid-tail and long-tail playlists, particularly when users spent more time at home, implies that listeners became more exploratory when their usual routines were changed. This pattern supports earlier work on how digital platforms with low discovery costs can enable broader content engagement (Brynjolfsson et al., 2010). For streaming platforms like Spotify, the growth in follower counts for mid-tail and long-tail playlists during periods of reduced mobility suggests that listeners were more willing to explore beyond the most-followed playlists. Rather than focusing solely on already-dominant playlists, platforms

could use this insight to adjust their recommendation algorithms, or featured categories to surface mid-tier content that matches emerging listening routines, such as playlists for working from home, relaxation, or daily rituals. Doing so during periods of heightened residential time or strict policy restrictions may help retain user interest and foster deeper engagement, without depending entirely on superstar playlists whose visibility is already saturated.

For other content curators, our findings suggest that relying solely on flagship playlists or label-driven branding may be less effective during these periods. As users shift their attention based on changing contexts (e.g., more time at home), curators may benefit from diversifying the types of playlists they maintain, including more thematic, mood-based, or situational collections that resonate with listeners' daily activities. Additionally, rather than focusing only on playlist ownership, curators should aim to increase track-level presence across multiple playlists, as our results show that track popularity was a stronger driver of follower growth than label affiliation alone.

For track producers, including artists, independent labels, and major record companies, these results highlight the importance of widespread track inclusion across diverse playlists. Visibility was more strongly associated with track presence than with brand cues, suggesting that investing in track-level distribution strategies across playlists of different sizes and curators can increase exposure. Especially during periods of disrupted routines, maximizing reach through broader playlist placement may offer better engagement outcomes than relying on brand reputation or single high-profile placements.

Limitations and Future Research

This study opens up several avenues for future research that could address some limitations in our analysis. While we focused on a 49-week period to capture immediate pandemic effects, an extended timeframe would allow researchers to observe whether changes in playlist engagement persisted, diminished, or transformed as restrictions evolved or lifted. Future research could extend the analysis to cover post-pandemic recovery periods or subsequent waves of restrictions to evaluate whether shifts in follower behavior persisted or returned to pre-pandemic patterns. It is also possible that changes in listening behavior during this period were driven in part by a shift in attention from music playlists to other content formats, such as podcasts, which gained traction during the pandemic. Future research could examine whether rising podcast consumption contributed to flat or declining playlist follower growth, providing a fuller picture of how streaming preferences evolved under changing daily routines.

Second, although we focus on global playlists, this approach limits our ability to examine geographic variation in country-level user engagement. Because our data do not include country-level playlist identifiers, we rely on weighted indices of government restrictions and residential mobility. While this provides an indirect way to account for international variation, future research could benefit from more granular data that allows for regional comparisons. Country-specific studies could examine how national policies, cultural preferences, or local promotion strategies

influence playlist engagement.

Third, our sample is restricted to playlists that existed before the pandemic declaration. While this choice allows us to track how established playlists responded to pandemic-related variations, it excludes newly launched playlists that may have been explicitly designed to meet emerging user needs, such as playlists for working from home, relaxation, or virtual gatherings. Future work could incorporate these newer playlists to understand content innovation and curation strategies during these periods.

Additionally, future research could explore how genre, mood, or activity-based themes influenced follower engagement independently of curator type or popularity tier. For example, playlists centered on emotional or motivational themes may have resonated more strongly with listeners navigating prolonged periods of isolation. Similarly, the role of personalized (algorithmically generated or user-specific) playlists deserves further attention, as such formats may engage users differently than public, platform-curated lists, particularly in the context of reduced physical mobility and increased screen time. Researchers might also consider experimental or simulation-based approaches to construct counterfactual scenarios, e.g., what platform engagement via playlists might have looked like without the pandemic. These designs could provide richer causal insights into how external shocks interact with platform dynamics and user behavior, beyond what observational data alone can reveal.

Lastly, future research could explore revenue-related impacts more directly, potentially focusing on how changes in engagement influence revenue distribution across labels and artists. Since revenue allocation on streaming platforms is often complex and not always transparent, examining metrics that are closely linked to artist and label income would enrich the understanding of how engagement trends during disruptions may translate into financial outcomes. Such research could ultimately guide artists, labels, and streaming platforms in aligning their strategies with user preferences and preparing for revenue fluctuations in times of restricted mobility and changing routines.

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